Trading App: summary and next steps

What is built

- **Live market data.** The last 1,000 candles load from Binance's public API, then a WebSocket keeps the newest candle live. No account or API key is needed.
- **Candlestick chart.** Candles with volume bars, a last-price line, price and time axes, autoscale, and 9 intervals from 1 minute to 1 month.
- **Touch gestures.** Pan with fling, pinch zoom, price drag and price-axis scaling, double-tap to autoscale, and long-press for a crosshair with an open, high, low and close legend.
- **Drawing tools.** Horizontal and trend lines that you can place, select, move, reshape and delete. They stay on their candles through zoom, scroll and interval changes.
- **Fast rendering.** The chart follows the Lightweight Charts model: a time scale and a price scale hold the view, and painters redraw from them in separate layers.

2.1 ms
to draw the first frame

Under 0.4 ms
per frame while panning or zooming

0
widget rebuilds while panning, zooming or using the crosshair

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features do the same work as Lightweight Charts

What can be done next

Performance	Why it helps
Update live candles without a rebuild	Each live tick still rebuilds about 19 widgets and copies all 1,000 candles. Passing candles to the painters through a notifier, and updating the last candle in place, would make a tick repaint only, as in Lightweight Charts.
Stop rebuilding on drawing drags	Each drag step rebuilds the gesture handler. Rebuilding it only when the drawing tool changes would make dragging a line repaint only.
Cache axis labels	About 20–25 labels are laid out again on every pan or zoom frame. Caching them by their text would cut most of the axis painting.

Reliability	Why it helps
Reconnect the live stream	If the WebSocket drops, the chart stops updating and doesn't try again. Reconnecting with a backoff, and refetching any missed candles, would keep it live.
Retry after an error	A failed load only shows a message. A retry button would save leaving and reopening the chart.

Features	Why it helps
More markets	Only BTC/USDT can be opened. A symbol search or watchlist would cover other Binance pairs.

Older history on scroll	Scrolling stops at the oldest of the 1,000 candles. Loading older pages as you reach the left edge would allow unlimited history.
Save drawings	Drawings are lost when you leave the chart screen. Saving them per symbol on the device would keep them.
Indicators	Moving averages, RSI, MACD or Bollinger Bands, drawn as extra painters on the same scales.
More drawing tools	Rays, rectangles, Fibonacci retracements and text notes, built on the same drawing model.
Mouse and trackpad	The chart responds to touch only. Mouse-wheel zoom, drag and hover would make it usable on desktop and web.
Dark theme	The app is light-only. A dark theme suits a trading screen.